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MILITARY SPECIFICATION

BONDING OPTICAL ASSEMBLIES (GLASS-TO-METAL)

1. SCOPE

1.1 This specification covers the bonding of optical assemblies (glass-to-metal) for use in military fire control instruments (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-268	Paper, Kraft, Untreated, Wrapping
MMM-A-131	Adhesives; Glass-to-Metal (For Bonding of Optical Elements)

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of

STANDARDS

Military

MIL-STD-810	Environmental Test Methods

(Copies of specifications, standards and drawings required by suppliers in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 12GP

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3. REQUIREMENTS

3.1 Assembly components.

3.1.1 Optical.- The optical component shall comply with requirements specified on the applicable drawing.

3.1.2 Metal.- The metal component shall comply with requirements specified on the applicable drawing.

3.1.3 Adhesive.- Bonding materials shall conform with Type I of MMM-A-131 (see 6.2)

3.2 Bonding process.- Unless otherwise specified by the procuring agency, assemblies shall be bonded in accordance with the procedure specified herein. Deviation from this procedure is prohibited unless approval is obtained from the contracting officer prior to use in production.

3.2.1 Cleaning.

3.2.1.1 Optical component.- Polishing compounds, pencil marks and other contaminants shall be removed from the surface of the glass to be bonded. The bonding surface shall be cleaned by scouring with a paste of pumice or calcium carbonate and water applied with a cotton swab. Residue shall be removed by rinsing with distilled water and wiping with a clean, dry cheesecloth followed by an ethyl alcohol rinse and wiping.

3.2.1.2 Metal component.- The metal bonding surface shall be of virgin metal free from corrosion products, burrs, surface impregnants, anodic, passivation, oxide and other surface treatments as well as any other surface contaminants; such imperfections should be removed by carefully rubbing with 1-G emery polishing paper while at the same time not destroying the physical dimensions and flatness of the bonding surface. The surface shall be wiped with ethyl alcohol-saturated cheesecloth and allowed to dry.

CAUTION: Clean surfaces are essential prerequisites for obtaining a satisfactory bond.

3.2.2 Bonding.- Bonding shall conform to the procedures outlined below. These techniques involve the basic operations for brush application of the two different liquid adhesives specified in MMM-A-131 as Type I, Class 1 and Class 2, on both the glass and metal surfaces to be bonded.

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3.2.2.1 Application of Class 1 adhesive.- Class 1 adhesive (solids content 8 ± 2%) shall be applied as follows:

a. Apply a coat of Class 1 adhesive to each of the bonding surfaces (glass and metal). A coat consists of a double pass with the brush over the entire bonding area.

b. Air dry for at least 5 minutes to evaporate solvent. Inspect visually; if adhesive is still moist, extend drying time sufficiently to dry completely.

CAUTION: Shortcutting drying time will result in unsatisfactory bonds.

c. Repeat steps "a" and "b" above until layer of adhesive has accumulated to a thickness of 0.005 to 0.006 inch (approximately 15 coats) on each surface.

CAUTION: The proper thickness of the Class 1 adhesive is critical to a satisfactory bond and shall be measured by a dial indicator or other suitable measuring device.

d. Air dry final coating on components for 1 hour at room temperature (60° to 90°F); then oven dry for 35 ± 5 minutes at 160° ± 5°F. Upon completion remove components from oven and cool to room temperature. In lieu of oven drying, an air dry for not less than 12 hours at room temperature to allow evaporation of solvents is permitted.

3.2.2.2 Application of Class 2 adhesive.- Class 2 adhesive (solids content 30 ± 30%) shall be applied over the baked Class 1 adhesive as follows:

a. Apply a coat of Class 2 adhesive over the final Class 1 adhesive coating on both components. A coat consists of a double pass with the brush over the entire bonding area.

b. Air dry for at least 5 minutes to evaporate solvent. Inspect visually; if adhesive is still moist, extend drying time sufficiently to dry completely.

CAUTION: Shortcutting drying time will result in unsatisfactory bonds.

c. Repeat steps "a" and "b" above until a layer of Class 2 adhesive has accumulated to a thickness of 0.002 to 0.003 inch (approximately 8 coats) on each surface.

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CAUTION: The proper thickness of the Class 2 adhesive is critical to a satisfactory bond and shall be measured by a dial indicator or other suitable measuring device.

d. Air dry final coating on components for 1 hour at room temperature and bake for 35 ± 5 minutes at 210°F; then, to preclude glass cracking, cool slowly in oven to 160° ± 5°F after which they should be removed and cooled to room temperature. Cooling may be accomplished by normal heat loss of oven after shutting off heat. In lieu of oven drying, an air dry for not less than 12 hours at room temperature to allow evaporation of solvents is permitted.

3.2.2.3 Assembly and curing.- The optical and metal components, coated with Class 1 and Class 2 adhesive, shall be assembled and cured as follows:

a. Final cleaning.- Immediately before positioning parts in bonding fixture, remove dust or loose particles from the adhesive surfaces with a watchmaker's blower or other suitable low pressure method using clean dry air. Do not brush, wipe or contact any part of these surfaces with the hand, cloths, solvents, cleaners, etc.

b. Positioning components in bonding fixture.

(1) The bonding fixture shall be designed to provide mechanical positioning of components to be bonded to an accuracy compatible with the applicable drawing and to maintain a uniform pressure by spring loading, with single point application at the center of the bonding area. Sufficient support shall be provided to protect components of the bonded assembly without straining or binding. Since the fixture, optical and metal components and the adhesives are influenced by heat during the curing cycle, care shall be exercised to fabricate fixtures from materials which will maintain alignment and pressures. All springs shall be thermo-mechanically stabilized by conditioning for 3 hours at 300° ± 5°F maintaining a 25 ± 2 psi pressure.

(2) Following final cleaning, place components for bonding in the fixture, maintaining proper alignment during the bonding and curing cycle. A single sheet of kraft paper conforming to UU-P-268, Grade B-No. 2, 50 weight shall be placed between the polished surfaces of the optical component and the fixture to protect critical surfaces from being scratched or damaged.

c. Application of pressure.- Maintain a pressure of 20-25 psi distributed evenly over the bonded area specified on the applicable drawing. (Pressure beyond bonding area creates strains which may result in distortion or separation subsequent to curing. Even

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distribution is particularly important where component is aluminized glass since any shifting will mar the aluminum coating.)

d. Curing bonding adhesives.- Place fixture holding prepared assembly in an unheated circulating air-type oven, and bring oven to a curing level of 275° ± 5°F, maintaining this for 2.5 hours.

CAUTION: Glue line temperature and time duration is a critical process control.

e. Control of oven temperature.- The temperature rise to 275°F and subsequent cooling to 160°F temperature for an assembly shall be controlled to avoid introduction of strain or cracking to the glass, especially during cooling. As a matter of caution, the cooling rate must not exceed 40°F per hour. By controlling the heat input or output to the oven, a uniform rate of temperature change can be maintained throughout the entire cycle.

3.3 Performance requirements.- The bonded assembly shall show no voids or separations within the bonded area in excess of 10% (total) of that area and shall meet the alignment, optical quality and performance requirements of the applicable drawing after the following exposure:

a. Mechanical shock.- 250g applied parallel to the bonded surfaces, in such a direction to produce the greatest stress on the bond, repeated 3 times. The shock pulse shall have a wave form approximating a half sine wave with a minimum time duration of 1.5 milliseconds.

3.4 Bond life requirements.- Bonded optical assemblies that have successfully withstood the requirements of 3.3 shall be capable of exposure to the following requirements without failure:

3.4.1 Low temperature thermal stability.- Bonded optical assemblies shall show no evidence of failure and shall meet the alignment and optical performance requirements of the applicable drawing after exposure to a temperature of -80° ± 2°F for a period of 2 hours. Failure is defined as separation of the bonded components in excess of the 10% total area permitted in 3.3, or fracture, shatter or chipping of the glass component.

3.4.2 Shock - humidity - shock stability.- Subsequent to meeting the low temperature stability requirements, the bonded optical assembly shall be capable of exposure to the following conditions in sequence (a, b, c):

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a. Mechanical shock.- Unless otherwise specified on the applicable drawing, cumulative shocks consisting of 3 each at 300g, 350g, 400g, 450g, 500g, 550g and 600g applied parallel to the bonded surfaces in such a direction to produce the greates stress on the bond. The shock pulse shall have a wave form approximating a half sine wave. For 300g, 350g and 400g the shock pulse shall have a minimum time duration of 1.5 milliseconds; for 450g, 500g, 550g and 600g the shock pulse shall have a minimum time duratio of 0.5 milliseconds.

(NOTE: For example (3.4.2a only), a specified maximum of 500g (on drawing) requires a cumulative total of 15 shocks)

b. Temperature-humidity.- Saturated humid atmosphere of 160° ± 5°F for 24 hours.

c. Post-humidity mechanical shock.- Within 1 hour after temperature-humidity exposure, cumulative shocks as specified in 3.4.2a.

Failure is defined as the complete separation of the glass component from the metal component.

3.5 Workmanship.- Workmanship shall be of the highest quality consistent with the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Performance requirements.

4.2.1 Mechanical shock.- Each bonded assembly shall be tested to the requirements of 3.3a utilizing the shock test of Procedure I of Method 516 specified in MIL-STD-810 except that the pulse shape,

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peak value, time duration and direction of shock shall be as specified in 3.3a. The bonded assembly shall meet the requirements specified in 3.3. Subsequent to the shock test, the bonded assembly shall be subjected to the test in 4.2.2.

4.2.2 Optical performance.- Each bonded assembly shall meet all alignment and optical performance requirements as specified in 3.3, subsequent to the tests specified in 4.2.1.

4.3 Bond life test.- Two bonded assemblies or 5%, whichever is greater, periodically selected at random from each batch (see 6.3) of cured bonded assemblies shall meet the requirements specified in 3.4 utilizing the test methods specified in 4.3.1, 4.3.2, 4.3.3 and 4.3.4 in sequence. The failure of any bonded assembly shall be indicative of the quality of the bonded assemblies within that particular batch. All bonded assemblies shall be prepared in accordance with Section 3, and deviations therefrom shall not be permitted without prior approval through the contracting officer.

NOTE: Bonded assemblies subjected to the bond life test are considered degraded and shall not be assembled in an instrument.

4.3.1 Low temperature thermal stability.- The bonded sample shall be subjected to the requirement specified in 3.4.1. The testing equipment utilized for this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test chamber shall be at room temperature when the sample is inserted. The temperature of the test chamber shall then be set to 0°F, and the bonded assembly shall be allowed to come to equilibrium at this temperature. From 0°F to -80°F the temperature of the test chamber shall be decreased at the step rate of 20°F per hour; that is, the chamber temperature shall be set to -20°F and allowed to remain at that temperature for 1 hour, then the temperature shall be set to -40°F and allowed to remain at that temperature for 1 hour, etc. After the bonded assembly stabilizes at -80°F, it shall be allowed to soak at that temperature for 2 hours. The above procedure shall then be reversed and the bonded assembly returned to room temperature. The bonded assembly shall meet the requirements specified in 3.4.1.

4.3.2 Mechanical shock.- The bonded assembly or assemblies shall be subjected to the shock requirements of 3.4.2a utilizing the shock test of Procedure I of Method 516 specified in MIL-STD-810 except that the pulse shape, peak value, time duration and direction of shock shall be as specified in 3.4.2a. Upon satisfactory completion of the shock tests, the bonded sample assembly or assemblies shall be subjected to the test specified in 4.3.3.

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4.3.3 Temperature-humidity.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The bonded sample assembly or assemblies shall be exposed to a temperature-humidity atmosphere as specified in 3.4.2b. Subsequent to the temperature exposure, the bonded sample assembly or assemblies shall then be subjected to the test specified in 4.3.4.

4.3.4 Post humidity mechanical shock.- The bonded assembly or assemblies shall be subjected to a shock test in accordance with 3.4.2a within the time limit specified in 3.4.2c utilizing the shock test of Procedure I of Method 516 specified in MIL-STD-810 except that the pulse shape, peak value, time duration and direction of shock shall be as specified in 3.4.2a. The bonded assembly or assemblies shall meet the requirements specified in 3.4.2.

4.4 Workmanship.- The bonded optics shall meet the requirements specified in 3.4. Failure of any requirement shall be cause for rejection and disapproval of the production materials and process control methods for that batch.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking are not applicable to this specification.

6. NOTES

6.1 Intended use.- Bonded optical assemblies are utilized in optical fire control instruments which are subject to gun fire shock, extremes of temperature, and humidity and other environmental conditions throughout the world. Optical assemblies fabricated with optical cement MIL-A-3920 are not covered by this specification.

6.2 Unacceptable bond.- Type II, dry film adhesive of MMM-A-131 is not acceptable for the intended use.

6.3 Definition of batch.- A batch is the quantity of identical bonded assemblies which have been prepared for curing and cured using the same materials, procedures and curing equipment (oven) during one calendar month.

6.4 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

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Preparing activity:

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Project No. 12GP-A029Y

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